

DATA CLEANING AND PREPARATION REPORT

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- Introduction

Data cleaning and preparation are essential steps in the data analysis process, as they help improve the accuracy, consistency, and reliability of a dataset before analysis is performed. This report presents the data cleaning procedures carried out on an e-commerce sales dataset containing customer transactions and sales information. Although the dataset was generally well structured, several quality checks and standardisation procedures were performed to ensure that the data was suitable for further analysis.

OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.1
ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral	302.7
ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	2753.4
ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	273.19
ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04
ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram	491.72
ORD200006	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	SAVE10	Facebook	664.42
ORD200007	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FREESHIP	Facebook	747.75
ORD200008	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692	2		Email	268.56
ORD200009	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	SAVE10	Google	2037.52
ORD200010	2023-12-29	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned	TRK98859248	9	WINTER15	Instagram	3129.85
ORD200011	2024-02-17	C93861	Monitor	3	49.14	831 Main St	Online	Returned	TRK48234646	7	SAVE10	Email	147.42
ORD200012	2024-10-15	C38785	Monitor	2	180.5	179 Main St	Debit Card	Pending	TRK20419991	6	FREESHIP	Referral	361
ORD200013	2023-08-30	C88348	Laptop	3	201.49	903 Main St	Credit Card	Returned	TRK81913874	4	FREESHIP	Facebook	604.47
ORD200014	2023-03-27	C98474	Tablet	2	393.33	980 Main St	Debit Card	Pending	TRK79186539	3	SAVE10	Instagram	786.66
ORD200015	2023-07-17	C39416	Printer	1	473.96	942 Main St	Cash	Delivered	TRK54930938	3		Google	473.96
ORD200016	2023-01-24	C28976	Printer	2	533.81	914 Main St	Credit Card	Pending	TRK13194259	3	FREESHIP	Email	1067.62
ORD200017	2024-03-02	C60672	Tablet	1	423.4	382 Main St	Debit Card	Shipped	TRK11972116	3		Referral	423.4
ORD200018	2025-03-26	C84329	Desk	1	431.44	891 Main St	Gift Card	Pending	TRK65191045	3	SAVE10	Facebook	431.44
ORD200019	2023-01-23	C14552	Monitor	5	224	273 Main St	Cash	Returned	TRK59101365	9	SAVE10	Google	1120
ORD200020	2023-05-08	C54829	Chair	5	62.51	587 Main St	Cash	Pending	TRK16718651	9	WINTER15	Instagram	312.55
ORD200021	2024-11-17	C99023	Monitor	4	342.95	569 Main St	Cash	Delivered	TRK52646983	6		Google	1371.8
ORD200022	2023-01-29	C34416	Monitor	2	49.62	633 Main St	Online	Pending	TRK75937350	2	SAVE10	Google	99.24
ORD200023	2025-04-09	C42372	Phone	4	273.25	755 Main St	Debit Card	Cancelled	TRK36168083	5	FREESHIP	Referral	1093
ORD200024	2024-10-30	C17470	Laptop	2	230.95	956 Main St	Online	Delivered	TRK28775057	3	WINTER15	Referral	461.9
ORD200025	2024-07-24	C42577	Monitor	1	399.98	709 Main St	Debit Card	Delivered	TRK69423327	3	WINTER15	Email	399.98
ORD200026	2024-03-06	C21944	Laptop	5	59.55	296 Main St	Online	Delivered	TRK80276812	6		Referral	297.75
ORD200027	2024-02-22	C97593	Printer	1	472.74	881 Main St	Online	Shipped	TRK89117573	6	WINTER15	Google	472.74
ORD200028	2023-11-21	C59823	Phone	4	160.07	568 Main St	Online	Pending	TRK65895539	5		Email	640.28
ORD200029	2024-09-20	C33260	Desk	3	423.53	930 Main St	Credit Card	Delivered	TRK59744090	6	SAVE10	Referral	1270.59
ORD200030	2025-03-22	C44267	Monitor	2	426.7	895 Main St	Online	Pending	TRK54751444	5	SAVE10	Referral	853.4
ORD200031	2023-06-30	C26445	Chair	3	500.08	996 Main St	Debit Card	Returned	TRK31229981	6		Facebook	1500.24
ORD200032	2023-05-01	C12388	Tablet	5	536.72	830 Main St	Cash	Delivered	TRK99261395	10		Email	2683.6
ORD200033	2023-06-05	C52184	Chair	1	162.77	912 Main St	Online	Cancelled	TRK73312471	4	WINTER15	Facebook	162.77
ORD200034	2023-01-07	C67195	Chair	4	576.52	926 Main St	Credit Card	Pending	TRK55467915	5	WINTER15	Facebook	2306.08
ORD200035	2024-12-07	C50304	Monitor	2	465.15	128 Main St	Online	Shipped	TRK46850888	2	SAVE10	Facebook	930.3

Figure 1: Original e-commerce sales dataset before data cleaning and preparation.

- Dataset Preparation

Prior to the data cleaning process, the dataset was converted into an Excel Table to improve data organisation and facilitate efficient data management. This structured format provided built-in sorting, filtering, and structured referencing features, which enhanced the accuracy and efficiency of subsequent data cleaning and validation activities.

- Data Cleaning Process

Following the initial preparation of the dataset, a series of data cleaning procedures were carried out to improve data quality and ensure the dataset was suitable for analysis. These

procedures included the identification and treatment of missing values, duplicate record verification, text standardisation, and data formatting and validation.

- **Missing Value Identification and Treatment**

The dataset was examined for missing values across all variables. Missing entries were identified in the CouponCode column. Rather than removing these records, the missing values were replaced with "NoCouponCode" to indicate that no promotional code was applied during the transaction. This approach preserved all observations while improving consistency within the dataset.

OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.1
ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral	302.7
ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	2753.4
ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	273.19
ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04
ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram	491.72
ORD200006	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	SAVE10	Facebook	664.42
ORD200007	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FREESHIP	Facebook	747.75
ORD200008	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692	2	NoCouponCode	Email	268.56
ORD200009	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	SAVE10	Google	2037.52
ORD200010	2023-12-29	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned	TRK98859248	9	WINTER15	Instagram	3129.85
ORD200011	2024-02-17	C93861	Monitor	3	49.14	831 Main St	Online	Returned	TRK48234646	7	SAVE10	Email	147.42
ORD200012	2024-10-15	C38785	Monitor	2	180.50	179 Main St	Debit Card	Pending	TRK20419991	6	FREESHIP	Referral	361
ORD200013	2023-08-30	C83348	Laptop	3	201.49	903 Main St	Credit Card	Returned	TRK81913874	4	FREESHIP	Facebook	604.47
ORD200014	2023-05-27	C98474	Tablet	2	393.33	980 Main St	Debit Card	Pending	TRK79186539	3	SAVE10	Instagram	786.66
ORD200015	2023-07-17	C39416	Printer	1	473.96	942 Main St	Cash	Delivered	TRK54930938	3	NoCouponCode	Google	473.96
ORD200016	2023-01-24	C28976	Printer	2	533.81	914 Main St	Credit Card	Pending	TRK13194259	3	FREESHIP	Email	1067.62
ORD200017	2024-03-02	C06072	Tablet	1	423.40	382 Main St	Debit Card	Shipped	TRK11972116	3	NoCouponCode	Referral	423.4
ORD200018	2023-03-26	C84329	Desk	1	431.44	891 Main St	Gift Card	Pending	TRK65191045	3	SAVE10	Facebook	431.44
ORD200019	2023-01-23	C14552	Monitor	5	224.00	273 Main St	Cash	Returned	TRK59101365	9	SAVE10	Google	1120
ORD200020	2023-05-08	C54829	Chair	5	62.51	587 Main St	Cash	Pending	TRK16718651	9	WINTER15	Instagram	312.55
ORD200021	2024-11-17	C99023	Monitor	4	342.95	569 Main St	Cash	Delivered	TRK52646983	6	NoCouponCode	Google	1371.8
ORD200022	2023-01-29	C34416	Monitor	2	49.62	633 Main St	Online	Pending	TRK75937350	2	SAVE10	Google	99.24
ORD200023	2025-04-09	C42372	Phone	4	273.25	755 Main St	Debit Card	Cancelled	TRK36168083	5	FREESHIP	Referral	1093
ORD200024	2024-10-30	C17470	Laptop	2	230.95	956 Main St	Online	Delivered	TRK28775057	3	WINTER15	Referral	461.9
ORD200025	2024-07-24	C42577	Monitor	1	399.98	709 Main St	Debit Card	Delivered	TRK69423327	3	WINTER15	Email	399.98
ORD200026	2024-03-06	C21944	Laptop	5	59.55	296 Main St	Online	Delivered	TRK80276812	6	NoCouponCode	Referral	297.75
ORD200027	2024-02-22	C97593	Printer	1	472.74	881 Main St	Online	Shipped	TRK89117573	6	WINTER15	Google	472.74
ORD200028	2023-11-21	C59823	Phone	4	160.07	568 Main St	Online	Pending	TRK65895539	5	NoCouponCode	Email	640.28
ORD200029	2024-09-20	C33260	Desk	3	423.53	930 Main St	Credit Card	Delivered	TRK59744090	6	SAVE10	Referral	1270.59
ORD200030	2025-03-22	C44267	Monitor	2	426.70	895 Main St	Online	Pending	TRK54751444	5	SAVE10	Referral	853.4
ORD200031	2023-06-30	C26445	Chair	3	500.08	996 Main St	Debit Card	Returned	TRK31229981	6	NoCouponCode	Facebook	1500.24
ORD200032	2023-05-01	C12388	Tablet	5	536.72	830 Main St	Cash	Delivered	TRK99261395	10	NoCouponCode	Email	2683.6
ORD200033	2023-06-05	C52184	Chair	1	162.77	912 Main St	Online	Cancelled	TRK73312471	4	WINTER15	Facebook	162.77
ORD200034	2023-01-07	C67195	Chair	4	576.52	926 Main St	Credit Card	Pending	TRK55467915	3	WINTER15	Facebook	2306.08
ORD200035	2024-12-07	C50304	Monitor	2	465.15	128 Main St	Online	Shipped	TRK46850888	2	SAVE10	Facebook	930.3
ORD200036	2025-05-08	C59337	Laptop	2	534.43	465 Main St	Debit Card	Pending	TRK50961250	5	NoCouponCode	Facebook	1068.86
ORD200037	2024-10-14	C10054	Laptop	2	76.74	409 Main St	Cash	Returned	TRK94005261	7	SAVE10	Instagram	153.48
ORD200038	2024-12-19	C15924	Laptop	1	245.89	549 Main St	Debit Card	Delivered	TRK14835975	3	SAVE10	Google	245.89
ORD200039	2025-02-27	C96654	Printer	3	256.46	666 Main St	Credit Card	Delivered	TRK85508960	4	SAVE10	Referral	769.38
ORD200040	2023-08-20	C84766	Desk	2	344.02	267 Main St	Online	Delivered	TRK78967923	4	FREESHIP	Instagram	688.04
ORD200041	2023-12-05	C20212	Monitor	3	611.45	657 Main St	Gift Card	Pending	TRK69733327	4	FREESHIP	Instagram	1834.35
ORD200042	2024-05-05	C28712	Printer	2	151.66	980 Main St	Cash	Cancelled	TRK40722558	6	NoCouponCode	Instagram	303.32

Figure 2: Missing coupon code values replaced with "NoCouponCode".

- **Duplicate Record Verification**

The dataset was checked for duplicate records and duplicate Order IDs to ensure that each transaction was uniquely represented. Duplicate checking was carried out using Excel's built-in data validation tools. The review confirmed that no duplicate records or duplicate Order IDs remained in the dataset, ensuring data integrity and supporting reliable analysis.

	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	Total
	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903		7 SAVE10	Instagram	
	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral	
	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	
	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	
	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	
	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram	
	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	SAVE10	Facebook	
	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FREESHIP	Facebook	
	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692	2	NoCouponCode	Email	
	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card			6	SAVE10	Google	
	2023-12-29	C43443	Tablet	5	625.97	333 Main St	Credit Card			9	WINTER15	Instagram	
	2024-02-17	C93861	Monitor	3	49.14	831 Main St	Online			7	SAVE10	Email	
	2024-10-15	C38785	Monitor	2	180.50	179 Main St	Debit Card			6	FREESHIP	Referral	
	2023-08-30	C88348	Laptop	3	201.49	903 Main St	Credit Card			4	FREESHIP	Facebook	
	2023-03-27	C98474	Tablet	2	393.33	980 Main St	Debit Card			3	SAVE10	Instagram	
	2023-07-17	C39416	Printer	1	473.96	942 Main St	Cash			3	NoCouponCode	Google	
	2023-01-24	C28976	Printer	2	533.81	914 Main St	Credit Card			3	FREESHIP	Email	
	2024-03-02	C60672	Tablet	1	423.40	382 Main St	Debit Card			3	NoCouponCode	Referral	
	2025-03-26	C84329	Desk	1	431.44	891 Main St	Gift Card	Pending	TRK65191045	3	SAVE10	Facebook	
	2023-01-23	C14552	Monitor	5	224.00	273 Main St	Cash	Returned	TRK59101365	9	SAVE10	Google	
	2023-05-08	C54829	Chair	5	62.51	587 Main St	Cash	Pending	TRK16718651	9	WINTER15	Instagram	
	2023-11-17	C99023	Monitor	4	342.95	569 Main St	Cash	Delivered	TRK52646983	6	NoCouponCode	Google	
	2024-01-29	C34416	Monitor	2	49.62	633 Main St	Online	Pending	TRK75937350	2	SAVE10	Google	
	2025-04-09	C42372	Phone	4	273.25	755 Main St	Debit Card	Cancelled	TRK36168083	5	FREESHIP	Referral	
	2024-10-30	C17470	Laptop	2	230.95	956 Main St	Online	Delivered	TRK28775057	3	WINTER15	Referral	
	2024-07-24	C42577	Monitor	1	399.98	709 Main St	Debit Card	Delivered	TRK69423327	3	WINTER15	Email	
	2024-03-06	C21944	Laptop	5	59.55	296 Main St	Online	Delivered	TRK80276812	6	NoCouponCode	Referral	
	2024-02-22	C97593	Printer	1	472.74	881 Main St	Online	Shipped	TRK89117573	6	WINTER15	Google	
	2023-11-21	C59823	Phone	4	160.07	568 Main St	Online	Pending	TRK65895539	5	NoCouponCode	Email	
	2024-09-20	C33260	Desk	3	423.53	930 Main St	Credit Card	Delivered	TRK59744090	6	SAVE10	Referral	
	2025-03-22	C44267	Monitor	2	426.70	895 Main St	Online	Pending	TRK54751444	5	SAVE10	Referral	
	2023-06-30	C26445	Chair	3	500.08	996 Main St	Debit Card	Returned	TRK31229981	6	NoCouponCode	Facebook	
	2023-05-01	C12388	Tablet	5	536.72	830 Main St	Cash	Delivered	TRK99261395	10	NoCouponCode	Email	
	2023-06-05	C52184	Chair	1	162.77	912 Main St	Online	Cancelled	TRK73312471	4	WINTER15	Facebook	
	2023-01-07	C67195	Chair	4	576.52	926 Main St	Credit Card	Pending	TRK55467915	5	WINTER15	Facebook	
	2024-12-07	C50304	Monitor	2	465.15	128 Main St	Online	Shipped	TRK46850888	2	SAVE10	Facebook	
	2025-05-08	C59337	Laptop	2	534.43	465 Main St	Debit Card	Pending	TRK50961250	5	NoCouponCode	Facebook	
	2024-10-14	C10054	Lantron	2	76.74	409 Main St	Cash	Returned	TRK94005761	7	SAVE10	Instagram	

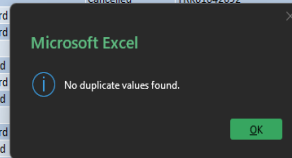


Figure 3: Duplicate record verification process.

- Text Standardisation

Text fields within the dataset were reviewed to identify inconsistencies that could affect the accuracy of the analysis, such as unnecessary spaces and variations in text formatting. Excel functions including TRIM and PROPER were applied where appropriate to remove extra spaces and standardise the presentation of text values. These checks helped improve the consistency of categorical variables and reduced the likelihood of duplicate categories being created due to formatting differences. The review confirmed that the text fields were appropriately structured and suitable for further analysis.

- Data Formatting and Validation

The dataset was examined to ensure that date, numerical, and categorical variables were consistently formatted throughout. Following its conversion into an Excel Table, the structured dataset facilitated efficient sorting, filtering, and validation of records. Date fields were validated to maintain a uniform format, while numerical variables such as Quantity, UnitPrice, ItemsInCart, and TotalPrice were reviewed to confirm that they were stored and displayed correctly. Additional validation checks were carried out to identify any obvious inconsistencies or data entry errors. The results confirmed that the dataset was properly formatted and ready for exploratory data analysis.

- **Conclusion**

The dataset was examined to ensure that date, numerical, and categorical variables were consistently formatted throughout. Following its conversion into an Excel Table, the structured dataset facilitated efficient sorting, filtering, and validation of records. Date fields were validated to maintain a uniform format, while numerical variables such as Quantity, UnitPrice, ItemsInCart, and TotalPrice were reviewed to confirm that they were stored and displayed correctly. Additional validation checks were carried out to identify any obvious inconsistencies or data entry errors. The results confirmed that the dataset was properly formatted and ready for exploratory data analysis.